

Product Carbon Footprint Study Report

Product Name: transistor production, surface-mounted

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: transistor production, surface-mounted

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg transistor production, surface-mounted

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg transistor production, surface-mounted.

2. System Boundary

transistor production, surface-mounted, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of transistor production, surface-mounted produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **24.664 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

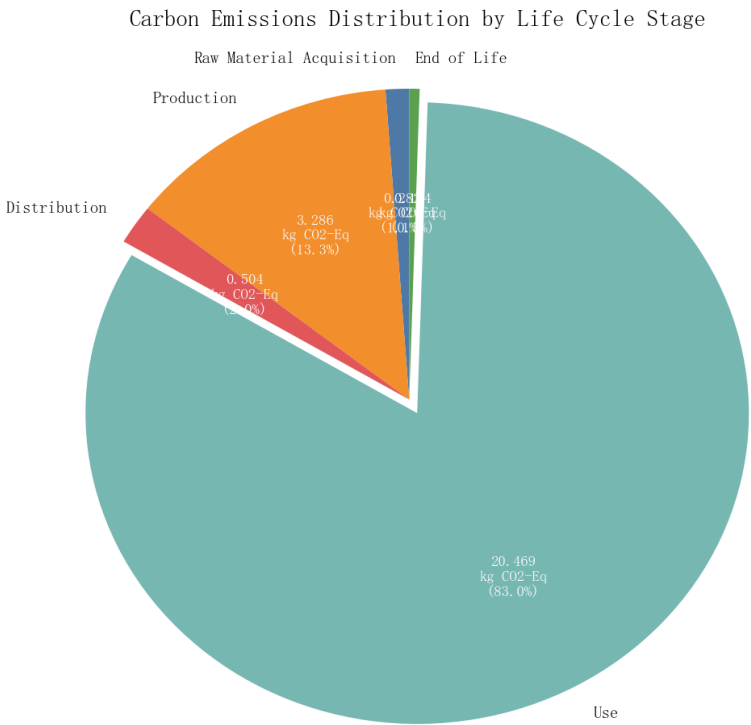


Table 1: Carbon Emissions by Life Cycle Stage

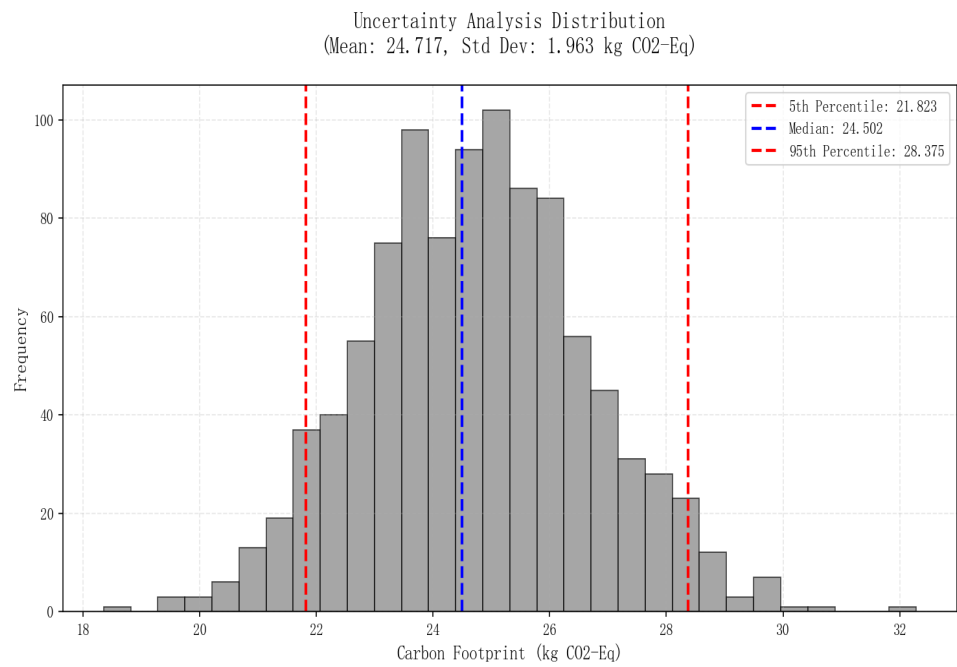
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.282	1.14
Production	3.286	13.32
Distribution	0.504	2.04

Use	20.469	82.99
End of Life	0.124	0.50
Total	24.664	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	24.717
Standard Deviation	1.963
Coefficient of Variation (CV)	7.94%
Median	24.502
5th Percentile	21.823
95th Percentile	28.375

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the surface-mounted transistor production is quantified at 24.664 kg CO₂-Eq. This figure reflects the aggregated greenhouse gas emissions across the product's full lifecycle, indicating a moderate-to-high impact level for an electronic component, primarily driven by energy-intensive phases rather than material intensity.

Lifecycle stage contributions exhibit a highly skewed distribution, with the use phase accounting for 83.0% (20.469 kg CO₂-Eq) of total emissions. Production follows at 13.3% (3.286 kg CO₂-Eq), while raw material extraction (1.1%), distribution (2.0%), and end-of-life treatment (0.5%) remain marginal. This pattern underscores that operational electricity consumption during the product's active service life dominates the impact profile, typical for semiconductor devices integrated into power-consuming electronics.

Major emission sources are therefore concentrated in the use phase, stemming from prolonged energy demand in end-user applications, with manufacturing processes representing the secondary hotspot due to process heat and electricity requirements.

2. Statement of Assumptions and Limitations

Key assumptions in this study include average European electricity grid mixes for the use phase, standard operational lifetimes of 5-10 years for transistors in typical circuits, and allocation of impacts based on mass or economic value where multi-output processes occur. Transportation distances are modeled using generic logistics data, and end-of-life scenarios assume partial recycling rates consistent with WEEE directives.

Limitations arise from reliance on secondary databases such as Ecoinvent, which may not fully capture site-specific production technologies or regional variations in energy sources. Data quality for foreground processes is often medium, with potential uncertainties in emission factors for specialty chemicals used in semiconductor fabrication. Temporal mismatches between dataset collection years and current practices further introduce variability.

System boundary restrictions encompass a cradle-to-grave scope but exclude indirect land-use changes, infrastructure construction, and downstream product assembly impacts. Use-phase modeling is bounded by assumed duty cycles without accounting for variable load profiles or efficiency improvements in modern circuits.

3. Suggestions for Improvement

- Optimize use-phase efficiency by redesigning transistors for lower power dissipation (e.g., via advanced doping or gate materials) to directly cut the 83% dominant emissions.
- Shift production electricity sourcing to renewable grids or implement on-site solar to reduce the 13.3% manufacturing contribution.
- Enhance raw material traceability and substitute high-impact precursors with lower-carbon alternatives where technically feasible.

- Improve end-of-life recovery through design-for-disassembly features to increase recycling yields beyond current 0.5% impact.
- Conduct primary data collection via supplier audits and real-time energy metering to elevate foreground data quality from medium to high precision.

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